

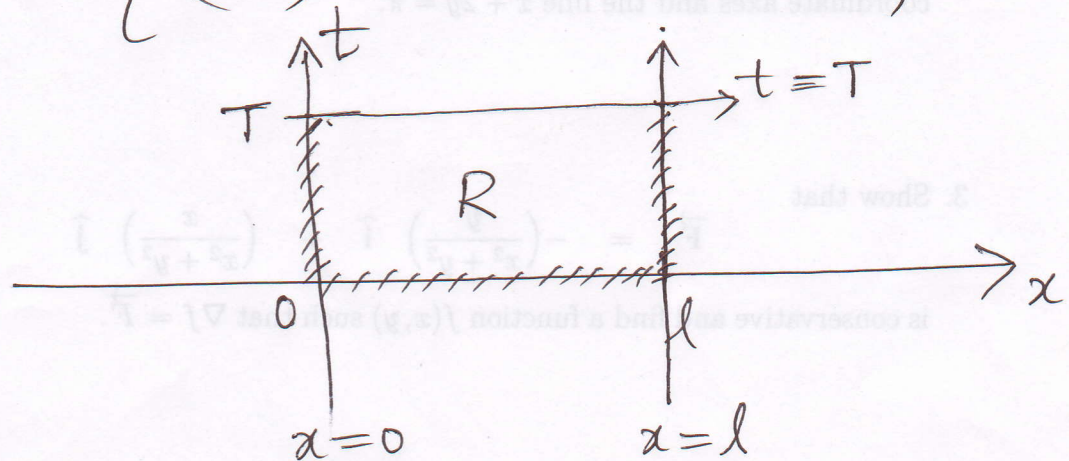
One dimensional heat Equation

①

$$\text{For } k > 0, \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < l, \quad 0 < t < T.$$

Define Rectangle R as follows:

$$R \equiv \{ (x, t) \in \mathbb{R}^2 : 0 \leq x \leq l, 0 \leq t \leq T \}$$



Consider the three sides of the rectangle R as $\Gamma \equiv \{ (x, t) \in R : t=0 \text{ or } x=0 \text{ or } x=l \}$.

Maximum Principle: If $u(x, t)$ satisfies the heat equation inside R then

$$\max_{\Gamma} u = \max_R u.$$

(2)

Proof: Suppose u attains its maximum at an interior point $\uparrow$ of R .

$$\Rightarrow u_x(x_0, t_0) = 0, \quad u_t(x_0, t_0) = 0$$

and $u_{xx}(x_0, t_0) \leq 0$.

Case 1: $u_{xx}(x_0, t_0) < 0$.

$$u_t(x_0, t_0) - k u_{xx}(x_0, t_0)$$

$$= -k u_{xx}(x_0, t_0)$$

$$> 0.$$

This contradicts the fact that u satisfies the heat equation at all points inside R .

Case 2: $u_{xx}(x_0, t_0) \leq 0$.

Define $v(x, t) \equiv u(x, t) + \epsilon x^2$, $\epsilon > 0$.

Since $\Gamma \subset R$

$$M \equiv \max_{\Gamma} u \leq \max_R u$$

To prove Maximum principle we only have to prove the reverse inequality:

(3)

To prove that $u(x,t) \leq M \quad \forall (x,t) \in R$.

$$\begin{aligned} \text{At points of } \Gamma, \quad v(x,t) &\leq \max_{\Gamma} v(x,t) \\ &= \max_{\Gamma} [u(x,t) + \epsilon x^2] \\ &= M + \epsilon l^2. \end{aligned}$$

If we prove $v(x,t) \leq M + \epsilon l^2$
for all points $(x,t) \in R$ we are
done because :

$$\begin{aligned} u(x,t) &= v(x,t) - \epsilon x^2 \\ &\leq M + \epsilon l^2 - \epsilon x^2 \\ &= M + \epsilon (l^2 - x^2) \end{aligned}$$

Choosing $\epsilon > 0$ as small as we please
we can achieve $u(x,t) \leq M$.

So let us prove the inequality
 $v(x,t) \leq M + \epsilon l^2 \quad \forall (x,t) \in R$.

(4)

Observe that $v_t - k v_{xx} < 0$:

$$v_t - k v_{xx} = u_t - k(u_{xx} + 2\epsilon)$$

$$= u_t - k u_{xx} - 2k\epsilon$$

$$= -2k\epsilon$$

$$< 0 \quad (\text{since } k > 0, \epsilon > 0).$$

Now in the following we prove that this new function $v(x, t)$ satisfies Maximum principle.

First ^{we prove that} $v(x, t)$ cannot attain its maximum inside R .

Suppose $(x_0, t_0) \in R$ be an interior point where $v(x, t)$ attains maximum.

Then $v_t(x_0, t_0) = 0$ and $v_{xx}(x_0, t_0) \leq 0$.

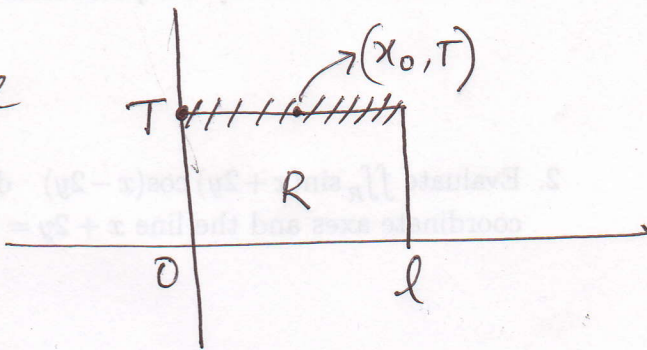
$$v_t(x_0, t_0) - k v_{xx}(x_0, t_0) = -k v_{xx}(x_0, t_0) \geq 0$$

This contradicts the above heat inequality $v_t - k v_{xx} < 0$.

Secondly we prove that $v(x,t)$ cannot attain its maximum on the fourth side of R (ie. $t=T$)

(5)

Suppose (x_0, T) be a point where $v(x,t)$ attains its maximum. Then



$$v_x(x_0, T) = 0, \quad v_{xx}(x_0, T) \leq 0$$

but $v_t(x_0, T)$ does not have to be zero.

$$v_t(x_0, T) \equiv \lim_{\delta \rightarrow 0^+} \frac{v(x_0, T-\delta) - v(x_0, T)}{-\delta} \geq 0.$$

$$\text{Now } v_t(x_0, T) - k v_{xx}(x_0, T) \geq 0$$

This again contradicts the heat inequality $v_t - k v_{xx} < 0$.

Because $v(x,t)$ is a continuous function it should attain its maximum on Γ .

(6)

Hence
$$v(x, t) \leq \max_R v(x, t)$$

$$= \max_{\Gamma} v(\gamma, t)$$

$$\leq M + \epsilon l^2.$$

Therefore $u(x, t)$ satisfies the Maximum principle.

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